The 40 Hour Cabin: the case

A storage building you can live in, owned the way an RV is owned.

> **The objective: a small home a person can own outright for the price of a used car -- $12,036 for the building, $18,276 standing on its pad -- that moves with them, grows with them, and never becomes the thing that sinks them.**

WHO IT IS FOR

Veterans. 32,882 were homeless on one night in January 2024 (HUD). A small, squared-away space, maintained on a schedule, built by a crew: a life a lot of veterans already know. Designed to sit beside HUD-VASH, SSVF, VA Grant and Per Diem.

Single mothers. A home that does not eat the household: one price, paid once, no mortgage, running costs of $129 a year to heat and cool, and a building that moves when the job or the school does.

Craftsmen. A trade, not a product: split, cut on one jig, assemble, maintain. Every cut is published; a crew that builds one can build the next, and sell them.

College students. A room you own near campus, on a rented pad, that comes with you, or sells to the next student, when you graduate.

Bachelors. A first place of your own for the price of a used car, that grows a panel at a time instead of a mortgage at a time.

THE MODEL: A STORAGE BUILDING, BUT LIVABLE; OWNED LIKE AN RV

Strip a house down to what it cannot do without and you have a shell, a floor, and the services. Build the shell the way a storage building is built -- simple, standard, the same every time -- and put the services in one place, the way an RV does: a **utility core** that plugs into a **serviced pad**. You own the building; you rent the ground, or own it. When you move, the building can come with you: the shell in 4 slices, the core for $260. At 10 ft across the same parts make a 79 sq ft shed; at the standard size, 277 sq ft of home; at 30 ft, 707 sq ft.

HOW THE BENEFITS STACK

1. A frame	120 split-log members, 40 panels: a covered, shaped space from trees you may already have	**$692**
2. A dry room	the weather shell, in 4 slices that two people carry and snap together	**$3,957**
3. A home's services	the utility core: 5 services in one column, the RV's hookup built in; it lasts 25 years and moves for $260	**$5,847**
4. Your building, not a box	snap-in panels -- 12 types, 25 modules -- make it a home, a studio, a shop or a storage room	**$6,337**
5. Built for you	paid build labour, about 67 hours -- or your own	**$8,221**
6. A business behind it	overhead, a warranty reserve, and a margin printed on the invoice	**$12,036**
7. A place to stand	a serviced pad, at cost; or rent one from a host, like an RV site	**$18,276**

And on top, for nothing extra: 42% less outside skin than a box of the same floor; heating and cooling $129 a year against $220; 9,870 gallons of rain a year off its own seams at 40 in of rain; insulation quilted from recycled clothing, about 132 shirts a layer, R-1.6 each.

WHY IT SCALES

Simple. 120 members on two lengths, 40 flat panels from one jig; the zome from the same wedges takes one bevel, 22.5°, for every cut. Nothing that needs a specialist, a crane or a factory to start.

Stripped down. A shell, a core, panels. Everything that makes it a particular building plugs in, so the body never changes and the factory never retools.

Repeatable. The same parts list at every size, the same cuts every time; the frame clock is 37.5 hours, measured where it could be. A crew that builds one has the jigs for the next.

Universal. 15 buildings from one body, from $12,036: a home, a studio, a shop, a nursery, cold storage. Wherever a shed can stand, this can.

Modular. 12 snap-in panel types ($180 to $1,480) and 25 modules; the core moves to the next building. Upgrades are a panel at a time.

Open. Every drawing, cut and number is published with the code that computes it. Anyone can build one; every builder makes the method stronger and the network bigger.

WHY IT IS HARD TO FAIL

The floor is a building. If it never becomes anybody's home, it is still a dry, serviced building with a use -- storage, a shop, a studio. The worst case is a good shed.

Small stakes. $12,036 is 20% of what the same floor costs as a new site-built house ($60,107) and 50% of a new manufactured home. Nobody loses a house on it.

It keeps its value by moving. No land is tied up in it; it can be sold, moved or re-fitted. The core outlives the shell around it.

Cheap to run, cheap to start. $91 a year not spent on heating and cooling; the splitting keeps 89% of a log that might otherwise be firewood.

Each piece is human-sized. Every strut weighs 15–22 kg, weighed: one person lifts it. No crane, no lender, no specialist trade to begin.

The host side pays. A pad host's $10,000 pad repays in 13.8 months at $1,000 a month, before financing and tax (a scenario).

The money repays. The ask, $100,650, is the margin on 51 buildings.

WHAT COULD STILL GO WRONG -- AND WHAT IS DONE ABOUT IT

It is not yet rated. No engineer has signed the frame for wind and snow. That review is the first line of the ask.

Rules differ by place. Whether a building this size may be lived in, and on what kind of pad, is for each county's zoning and code to say. The small size is shed-sized; the standard size needs a permit in most places.

It is not yet measured over a year. Moisture in the seams and the skin's upkeep are modelled, not measured; the instrumented test dome is in the ask.

The work is real. About 67 hours for the whole building, not forty; the starter pad computes at $13,148, $3,148 over its target. These numbers are printed because they argue back.

THE ASK

$100,650, the sum of its costed lines: prove it ($12,850), make it repeatable ($60,800), build what an owner cannot ($22,000), seed the network ($5,000). The numbers page behind this one shows every line of the arithmetic. Built by Donovan Zeanah, a U.S. Navy veteran, in the open: github.com/dkzeanah/DomeSimulator.